

Chlorhexidine vs Povidone-Iodine Alcohol Solutions for Cardiac Implantable Electronic Devices



A Prospective Randomized Study

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ABSTRACT

BACKGROUND Cardiac resynchronization therapy (CRT) improves heart failure outcomes and reduces morbidity and mortality but carries the highest infection risk among cardiac implantable electronic devices. Studies on surgical-site infections show conflicting results regarding the use of alcohol-based povidone-iodine vs chlorhexidine for skin anti-sepsis, especially as CRT implantation remains insufficiently studied.

OBJECTIVES This study sought to compare the effectiveness of 2% alcohol-based chlorhexidine with that of 4-phase povidone-iodine-alcohol skin preparation in preventing local or systemic infections after cardiac resynchronization therapy device implantation, and to assess the impact on other major cardiovascular events and treatment-related adverse effects over 24 months.

METHODS In this randomized, multicenter, open-label trial with blinded outcome adjudication, patients undergoing CRT (implantation, upgrade, or replacement) pacemaker, or defibrillator implantation were assigned 1:1 to receive skin antisepsis with either 2% alcoholic chlorhexidine or alcoholic povidone-iodine. Randomization was conducted centrally using an interactive web-response system, with stratification according to trial center. The primary outcome was local or systemic infection related to the device over a 24-month follow-up period. The secondary outcomes were major cardiovascular events and noninfectious side effects.

RESULTS From April 2013 to December 2018, 2,272 patients were randomized to the chlorhexidine (n = 1,143) or the povidone-iodine group (n = 1,129). Median age was 72 years; 75.1% were men. Device-related infections occurred in 2.9% of the chlorhexidine group and 3.9% of the povidone-iodine group (adjusted subhazard ratio: 0.75; 95% CI: 0.48-1.20; P = 0.23). Major cardiovascular events occurred in 31.5% and 31.3% of patients, respectively (subhazard ratio: 1.01; 95% CI: 0.87-1.17). Noninfectious local side effects occurred in 12.9% vs 13.3%.

CONCLUSIONS The study did not demonstrate a significant reduction in infection rates with alcoholic chlorhexidine compared with alcoholic povidone-iodine in patients undergoing cardiac resynchronization device implantation. Both antiseptics showed similar rates of major infectious and noninfectious events. Further research is needed to optimize infection prevention strategies in this population. (Comparison of Alcoholic Chlorhexidine 2% Versus Alcoholic Povidone Iodine for Infections Prevention With Cardiac Resynchronization Therapy Device Implantation [CHLOVIS]; [NCT01841242](https://doi.org/10.1016/j.jacc.2025.10.042)) (JACC. 2026;87:200-211) © 2026 by the American College of Cardiology Foundation.



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Cardiac implantable electronic devices (CIEDs) carry their own set of risks and benefits, with infections remaining a major complication of permanent pacemaker and implantable cardioverter-defibrillator implantation.¹⁻⁵ CIED infection occurrence varies from 0.68% to 4.8%, depending on multiple factors, along with a 3-fold increased risk of CIED mortality.¹ Overall, 1-year mortality of CIED infection was about 16.9%.^{1,4} The national health care administrative databases in the United States reported an increased CIED infection ranging from 1.45% in 2000 to 3.41% in 2012, with higher-than-expected CIED infection rates, as illustrated by about 5% in the DANISH (Danish ICD Study in Patients With Dilated Cardiomyopathy) trial.^{2,5} Till now, several models have been published, designed to help estimate this infectious risk.^{6,7} Thus, the contemporary PADIT (Prior procedures, Age, Depressed renal function, Immunocompromised state, Type of procedure) score proves to be very useful for predicting the infection risk including technical factors, patient factors, periprocedural factors, and the device type procedure.^{6,7} PADIT score and

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BLISTER (Blood results [eg, eGFR, CRP], Long procedure time, Immunosuppressed, Sixty years old [or younger], Type of procedure, Early reintervention/Repeat procedure) score, and other studies proved that cardiac resynchronization therapy (CRT) carries the highest infectious risk due to the procedure complexity, including the number of the leads, the generator size, the procedure duration, and the patient comorbidities or frailties.⁶⁻⁹ Accordingly, as previously reported, the risk of CRT device infection is at least doubled or tripled compared with the standard permanent pacemaker or implantable cardioverter-defibrillator implantation risk.⁶⁻⁹

Moreover, in the CIEDs infectious field prevention, Da Costa et al¹⁰ demonstrated a long time ago that the preaxillary flora contamination during the procedure plays a key role in CIED infection mechanism. Considering that the preprocedure and intraprocedure contamination represents the CIED infection's main mechanism, all our efforts should be concentrated on local and general prevention. In a meta-analysis, we demonstrated the impact of intravenous systemic antibiotic prophylaxis 1 hour before the procedure, on reduction of CIED infection (–75%).¹¹ Preoperative antiseptic skin cleansing also prevents CIED infections by the elimination of colonizing bacteria and their potential role in CIED infection genesis, but the optimal topical antiseptic choice is still unclear.¹²⁻¹⁴ Although Darouiche et al¹² demonstrated that 2% chlorhexidine-alcohol was better than povidone-iodine for skin preparation before general surgery (patients undergoing abdominal, thoracic, gynecological, and urological surgery) or intravascular catheter insertion, few data, and no randomized study specifically related to CIED implantation procedures exist. Clinical practice guidelines universally recommend the use of antiseptic skin solutions containing alcohol to prevent surgical-site infection. Although, some guidelines favor antiseptic with chlorhexidine gluconate over an iodophor, all recommendations recognize a lack of consensus with respect to the most effective agent.⁴

Given the limited evidence of a benefit and no randomized study in this field of CIED antiseptic prevention, we conducted a randomized trial in which patients who had chronic symptomatic heart failure, reduced ejection fraction, and indication for CRT were randomly assigned to either receive povidone-iodine-alcohol solution or 2% chlorhexidine-alcohol.

ABBREVIATIONS AND ACRONYMS

CIED = cardiac implantable electronic device

CRT = cardiac resynchronization therapy

CRT-D = cardiac resynchronization therapy-defibrillator

CRT-P = cardiac resynchronization therapy-pacemaker

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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METHODS

TRIAL DESIGN AND OVERSIGHT. CHLOVIS (Comparison of Alcoholic Chlorhexidine 2% Versus Alcoholic Povidone Iodine for Infections Prevention With Cardiac Resynchronization Therapy Device Implantation; [NCT01841242](#)) was a French, prospective, multicenter, open-label study with blinded adjudication of outcome events, comparing 2% alcoholic chlorhexidine with alcoholic povidone-iodine. The trial was managed by the University Hospital of Saint-Étienne and supported by a grant from the French Ministry of Health through the National Hospital Clinical Research Program (PHRC-N 1108096). The protocol and amendments were approved by a central independent ethics committee (Ethic Committee of Sud Est I), and all patients provided written informed consent. It was coordinated by the Clinical Research Unit of the University Hospital of Saint-Étienne. Data were collected and maintained using the CleanWeb electronic data capture system and analyzed by the Clinical Investigation Center of the University Hospital of Saint-Étienne. A central adjudication committee, whose members were unaware of the treatment assignments, reviewed all suspected infections. A data safety monitoring board was organized, and interim analyses planned and executed. The trial investigators and committee members are listed in the [Supplemental Appendix](#).

PATIENTS. Patients requiring cardiac resynchronization therapy (NYHA functional class III–IV, ejection fraction $\leq 35\%$, wide left bundle branch block [≥ 130 ms], and optimal medical therapy)¹⁵; those requiring the implantation of a resynchronization device based on the PAVE (Post AV Nodal Ablation Evaluation) study (ejection fraction $\leq 45\%$ in atrial flutter or atrial fibrillation) and indication for atrioventricular node ablation; those requiring an implantable defibrillator or pacemaker but classified as NYHA functional class II (ejection fraction $\leq 35\%$, wide left or right bundle branch block [≥ 150 ms], and optimal medical treatment); those requiring an upgrade at least 2 years after their last implantation or those requiring a cardiac resynchronization device replacement >2 years after the previous implantation, were eligible to participate.¹⁶ When the resynchronization procedure failed, the implanted device was left to the operator choice and considered as other implantation device. All patients provided written informed consent. Exclusion criteria were as follows: history of allergy to chlorhexidine, alcohol, or iodophors; evidence of infection at or adjacent to

operating site; perceived inability to monitor the patient's postsurgery evolution, pregnant or breast-feeding women, and no written informed consent ([Supplemental Appendix](#)).

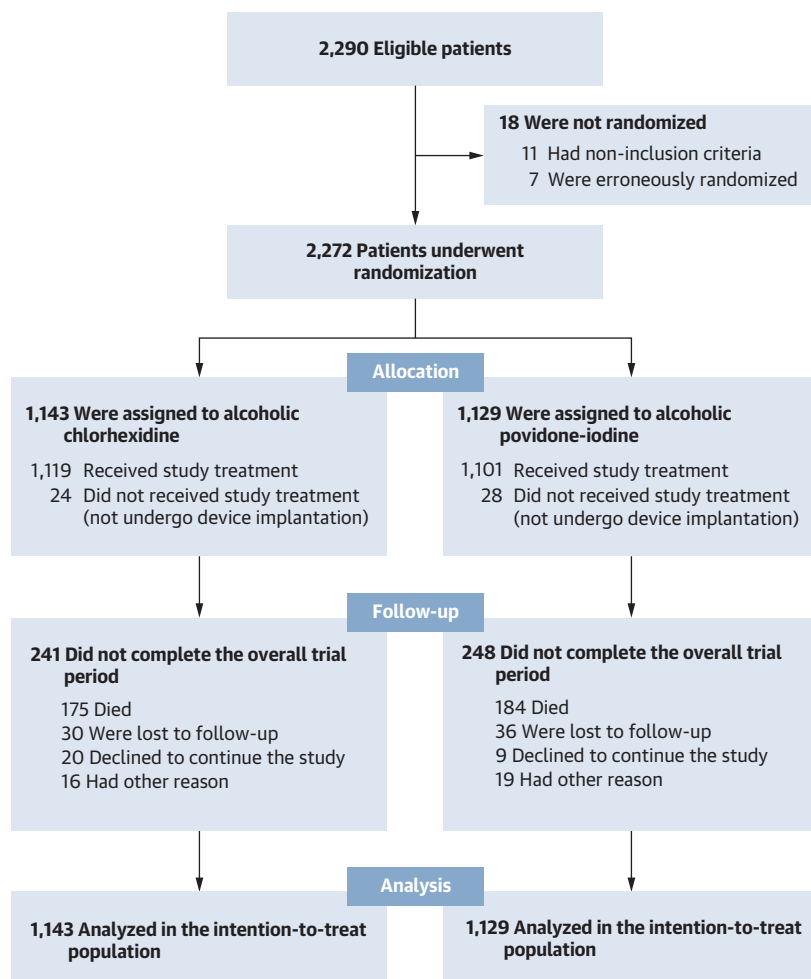
RANDOMIZATION AND TRIAL INTERVENTION. Patients were randomly assigned, in a 1:1 ratio, to the 2% alcoholic chlorhexidine group or the alcoholic povidone-iodine group. Randomization was conducted centrally using an interactive web-response system, with stratification according to trial center. The criteria for including patients in the intention-to-treat analysis included randomization and the possibility of applying each of the study antiseptic preparations (which required performance of surgery). All patients were systematically shaved and subjected to an antiseptic shower with either aqueous povidone-iodine 10% solution or Hibiscrub cutaneous solution (chlorhexidine gluconate 4%, Mölnlycke) on the night before the intervention. Antisepsis procedures were performed immediately before surgery, with 2 antiseptic solutions successively applied to the skin. Patients assigned to the alcoholic chlorhexidine group, the skin at the surgical site was preoperatively scrubbed with an applicator containing chlorhexidine gluconate 2% and isopropyl alcohol 70% (Chloraprep, Cardinal Health, 10.5 mL). The procedure was repeated twice. For those assigned to the alcoholic povidone-iodine group, skin preparation was performed by a registered nurse by applying first a 4% aqueous povidone-iodine solution (Scrub Care Skin Prep Tray, Cardinal Health) and then an alcoholic povidone-iodine 10% solution after rinsing and drying. Last, a povidone-iodine-alcohol 10% solution was applied twice by the surgeon. Moreover, each center received dedicated training on the 2 antiseptic protocols at the start of the study.

All patients received the same antibiotic prophylaxis. Intravenous administration of prophylactic antibiotic was performed 60 to 90 minutes before incision using a single cefazolin dose (1.5 g).^{11,17} When necessary, the dose was adjusted to the weight (<50 kg and ≥ 120 kg). In case of allergy, clindamycin or vancomycin were used depending on local policies. No local antibiotic pocket or saline washes were performed.

Trial visits were scheduled at enrollment, between 10 days and 3 months, then at 6, 12, 18, and 24 months after randomization. Additional visits were scheduled in the event of suspected infections.

OUTCOME MEASURES. The primary efficacy outcome was centrally adjudicated local or systemic infections related to the implantable device over the 24-month

FIGURE 1 Randomization, Treatment, and Follow-Up of the Patients



follow-up period. Infections were defined according to the criteria previously described.^{3,4,8-12,17,18} Clinical evidence included local inflammation signs at the generator pocket, such as erythema, warmth, fluctuance, wound dehiscence, tenderness, purulent drainage, or frank erosion caused by the generator or lead perforating the skin. Device-related endocarditis was clinically confirmed by detection of valvular or lead vegetations via echocardiography, or if the modified Duke criteria for infective endocarditis were met.^{3,18} CIED infection was microbiologically confirmed by positive culture results from samples of the generator pocket, leads, or blood, in the presence of local inflammatory signs at the generator pocket, absence of another bacteremia source, and resolution of bloodstream infection following device extraction.^{3,18} Once the infection was confirmed, patients required active antibiotic therapy and

system removal based on the international recommendations.^{3,4}

Secondary outcomes included major cardiovascular events and noninfectious side effects attributable to local treatment over the 24-month follow-up period. Major cardiovascular events were defined as a composite of heart failure (hospital admission for worsening heart failure), right heart embolic stroke, stroke of cardiac origin, myocardial infarction, other cardiovascular events, or death from any cause.

An additional exploratory post hoc analysis of the primary efficacy outcome (centrally adjudicated local or systemic infections related to the implantable device) was performed over the 12-month follow-up period.

STATISTICAL ANALYSIS. The estimated infection rate after cardiac resynchronization device implantation was 3.5% to 4% at 2 years.^{9,15} By analogy with

TABLE 1 Demographic and Clinical Characteristics at Baseline (Intention-to-Treat Population)

	Alcoholic Chlorhexidine (n = 1,143)	Alcoholic Povidone-Iodine (n = 1,129)
Age, y	71.6 ± 10.0	71.0 ± 11.0
Age, y		
<60	135 (11.8)	168 (14.9)
60-69	330 (28.9)	305 (27.0)
≥70	678 (59.3)	656 (58.1)
Male	853 (74.6)	853 (75.6)
Body weight, kg	78.5 ± 17.3	78.8 ± 16.7
Body mass index, kg/m ²	27.4 ± 5.2	27.3 ± 5.1
History of diabetes	366 (32.2)	352 (31.4)
History of immunosuppression	10 (0.9)	7 (0.6)
Antiplatelet therapy at inclusion ^a	545/1,135 (48.0)	511/1,120 (45.6)
Anticoagulant treatment at inclusion ^b	583/1,135 (51.4)	572/1,120 (51.1)
PADIT score ^c		
Low risk of infection	736 (64.4)	745 (66.0)
Moderate risk of infection	306 (26.8)	286 (25.3)
High risk of infection	40 (3.5)	37 (3.3)
Unavailable	61 (5.3)	61 (5.4)
Procedure duration, min	77.8 ± 40.1	78.8 ± 42.0
Fluoroscopy duration, min	16.1 ± 14.8	17.5 ± 16.3
Indications for implantation		
Primo cardiac resynchronization	798 (69.8)	807 (71.5)
Device upgrade	198 (17.3)	191 (16.9)
Cardiac resynchronization device replacement	146 (12.8)	130 (11.5)
Unknown	1 (0.1)	1 (0.1)
Type of implanted device		
Triple-chamber defibrillator	689 (60.3)	678 (61.8)
Biventricular defibrillator	108 (9.4)	111 (10.1)
Triple-chamber pacemaker	186 (16.3)	186 (17.0)
Biventricular pacemaker	76 (6.6)	68 (6.2)
Other	47 (4.1)	54 (4.8)
Procedure cancelled	24 (2.1)	28 (2.5)
Implantation failure	13 (1.1)	4 (0.3)

Values are mean ± SD, n (%), and n/N (%). Percentages may not total 100 because of rounding. ^aAntiplatelet therapy in the alcoholic chlorhexidine group vs alcoholic povidone-iodine were: aspirin (42.2% vs 40.5%), clopidogrel (12.9% vs 12.5%), ticagrelor (0.9% vs 1.1%), and prasugrel (0.3% vs 0.3%). ^bAnticoagulant treatment in the alcoholic chlorhexidine group vs alcoholic povidone-iodine were: vitamin K antagonist (35.4% vs 34.3%), direct oral anticoagulant (13.8% vs 14.5%), and heparin (3.3% vs 3.2%). ^cThe calculated score is a minimum estimate because data on depressed renal function and the number of prior procedures were not available in the database. Therefore, for depressed renal function, a score of 0 was assigned to all patients. For prior procedures, 0 point was assigned for primo cardiac resynchronization, and 1 point was assigned for device upgrade or replacement.

PADIT = Prior procedures, Age, Depressed renal function, Immunocompromised state, Type of procedure.

studies on general antibiotic prophylaxis or in this type of surgery, a 50% reduction in infection risk may be expected.^{11,12,17} Assuming an estimated infection incidence of 4% with alcoholic povidone-iodine and a 50% reduction in infections with alcoholic chlorhexidine, we estimated that 2,276 patients (1,138 patients per group) would be required to achieve 80% power to demonstrate the superiority of alcoholic chlorhexidine, with a 2-sided type I error of 5%. The study was prospectively powered to detect a 50%

difference. Interim analyses were planned and executed. The study was conducted until its terms.

To determine the superiority of alcoholic chlorhexidine vs alcoholic povidone-iodine, a time-to-event analysis was performed using the Fine and Gray regression model, with randomization strata as covariates to account for the competing risk of death¹⁹ in the intention to-treat population. The effect of the treatment was estimated using the sub-distribution HR and its 95% 2-sided CI, and overall significance was set at the 2-sided type I error of 0.05. Outcome analysis considered the time from randomization to the first event, accounting for the competing risk of death, or to the last known follow-up date if neither the event of interest nor a competing event occurred during 24-month follow-up (censored time). No imputation for missing data was performed, and patients' data were censored at the date of their last encounter. Descriptive subgroup analyses to assess homogeneity of the treatment effect were conducted for the primary outcome. Statistical analyses were performed using SAS software, version 9.4 (SAS Institute Inc).

RESULTS

PATIENTS AND TREATMENTS. The inclusion period extended from April 25, 2013, to December 18, 2018, during which time a total of 2,272 patients were randomized across 21 centers in France. Of these, 1,143 were assigned to the alcoholic chlorhexidine group and 1,129 to the alcoholic povidone-iodine group (Figure 1). All patients were followed for 24 months. The median follow-up duration was 24 months (Q1-Q3: 23 to 24.4 months). The baseline demographic and clinical characteristics of the patients were similar between the 2 groups (Table 1). The median age of the patients was 72 years (Q1-Q3: 65 to 79 years), and 1,706 patients (75.1%) were men. The indications for implantation were a primo cardiac resynchronization in 1,605 patients (70.6%), device upgrade in 389 patients (17.1%), and CRT generator replacement in 276 patients (12.2%). The indication was unknown in 2 patients (1 in each group). The indications for primo CIED implantation were 1,605 (70.6%): a primo cardiac resynchronization therapy-defibrillator (CRT-D) in 1,132 patients (70.5%), a primo cardiac resynchronization therapy-pacemaker (CRT-P) device in 359 patients (22.4%), and others in 114 (7.1%). When the resynchronization procedure failed, the implanted device was left to the operator choice. The indications for a CRT device upgrade were 389 (17.1%), CRT-D in 229 (58.9%), CRT-P in 118 (30.3%), and others in 42 patients

TABLE 2 Clinical Outcomes During the Trial Period

	Alcoholic Chlorhexidine (n = 1,143)	Alcoholic Povidone-Iodine (n = 1,129)	Treatment Effect (95% CI)	P Value
Primary outcome: local or systemic infection in relation to the implantable device	32 (2.9)	42 (3.9)	0.75 (0.48-1.20)	0.23
Secondary outcomes				
Major cardiovascular events	340 (31.5)	334 (31.3)	1.01 (0.87-1.17)	—
Heart failure	245 (22.9)	228 (21.3)	1.08 (0.90-1.29)	—
Right heart embolic stroke	0 (0.0)	0 (0.0)	—	—
Stroke of cardiac origin	17 (1.6)	16 (1.5)	1.05 (0.53-2.09)	—
Myocardial infarction	6 (0.6)	5 (0.5)	1.19 (0.36-3.89)	—
Other cardiovascular event	49 (4.5)	44 (4.0)	1.10 (0.73-1.65)	—
Death from any cause	175 (16.8)	184 (17.4)	0.93 (0.76-1.15)	—
Cause of death				
Cardiac	106 (60.6)	99 (53.8)	—	—
Infectious	34 (19.4)	42 (22.8)	—	—
Sudden death	9 (5.1)	9 (4.9)	—	—
Other cause	72 (41.1)	80 (43.5)	—	—
Hematoma	131 (11.5)	130 (11.5)	—	—
Noninfectious side effects attributable to local treatment	18 (1.6)	22 (1.9)	—	—
Urticaria/rash/pruritus	16 (1.4)	12 (1.1)	—	—
Contact eczema	1 (0.1)	6 (0.5)	—	—
Local burns	2 (0.2)	3 (0.3)	—	—
Angioedema, Quincke's edema	0 (0.0)	1 (0.1)	—	—
Caustic dermatitis	0 (0.0)	0 (0.0)	—	—

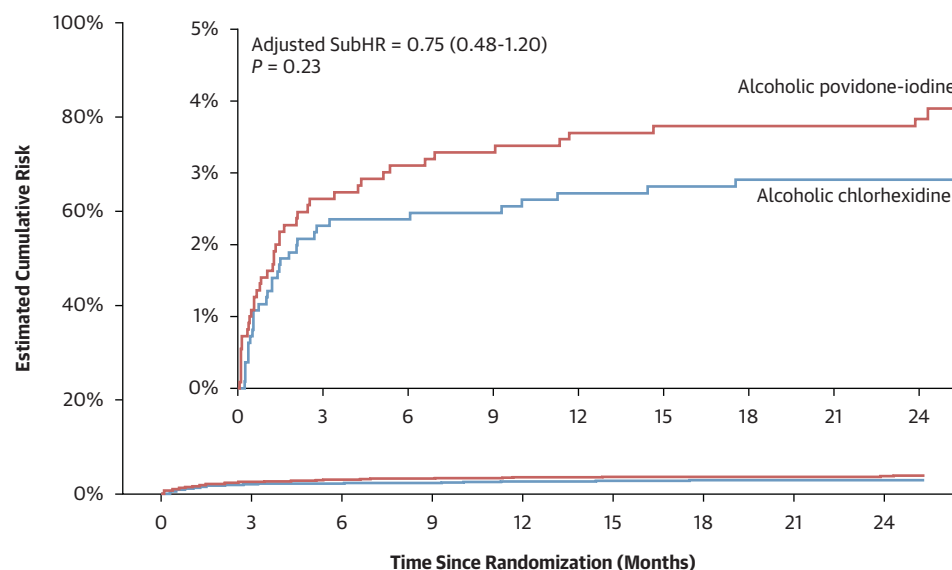
Values are n (%). Percentages are the cumulative incidence and thus may not calculate as expected, except for the analysis of side effects attributable to local treatment. The treatment effect was estimated as the subdistribution HR (subhazard ratio) with adjustment for the randomization strata, except for the analyses of death from any cause, for which a HR is reported.

(10.8%). The indications for device replacement were 276 (12.2%), a CRT-D generator replacement in 225 patients (81.5%), a CRT-P generator replacement in 39 patients (14.1%), and others in 12 patients (4.4%). “Others” represents the group where the resynchronization procedure failed or the device was not implanted or the procedure was cancelled or when the device was downgraded due to patient related conditions. Regarding the type of device implanted, a defibrillator was used in 1,586 patients (69.8%), a pacemaker in 516 patients (22.7%), and another type of device in 170 patients (7.5%).

PRIMARY OUTCOME. In the intention-to-treat analysis, local or systemic infection related to the implantable device occurred in 32 patients (cumulative incidence 2.9%) in the alcoholic chlorhexidine group and in 42 patients (cumulative incidence 3.9%) in the alcoholic povidone-iodine group (adjusted subhazard ratio: 0.75; 95% CI: 0.48-1.20; $P = 0.23$) (Table 2, Figure 2). This corresponds to an absolute risk difference of -1.0% (95% CI: -2.5% to 0.5%). Cumulative incidences of local infection were 2.6% and 2.9%, respectively. Although the primary endpoint was prespecified at 24 months, most events occurred

within the first 12 months (30 patients in the alcoholic chlorhexidine group and 39 patients in the alcoholic povidone-iodine group). An exploratory post hoc analysis at 12 months showed results consistent with the 24-month analysis (adjusted subhazard ratio: 0.77; 95% CI: 0.48-1.23), suggesting that the main findings are robust over time. Descriptive subgroup analyses of device-related infections did not show any apparent heterogeneity (Figure 3).

SECONDARY OUTCOMES. Major cardiovascular events occurred in 340 patients (cumulative incidence 31.5%) in the alcoholic chlorhexidine group and in 334 patients (cumulative incidence 31.3%) in the alcoholic povidone-iodine group (subhazard ratio: 1.01; 95% CI: 0.87-1.17) (Table 2). The most frequent major cardiovascular event was heart failure, occurring in 245 patients (cumulative incidence 22.9%) in the alcoholic chlorhexidine group and 228 patients (cumulative incidence 21.3%) in the alcoholic povidone-iodine group. Death from any cause occurred in 175 patients (cumulative incidence 16.8%) in the alcoholic chlorhexidine group and in 184 patients (cumulative incidence 17.4%) in the alcoholic povidone-iodine group (Table 2). Most

FIGURE 2 Local or Systemic Infections Related to the Implantable Device (Intention-to-Treat Population)

No. at Risk

Alcoholic Chlorhexidine	1,143	1,048	1,023	1,001	969	942	918	894	757
Alcoholic Povidone-Iodine	1,129	1,028	997	964	942	915	889	874	735

Shown is the cumulative incidence of local or systemic infections related to the implantable device (primary efficacy outcome) among patients who received alcoholic chlorhexidine or alcoholic povidone-iodine. The inset shows the same data on an expanded y-axis.

deaths were cardiac-related (60.1% in the alcoholic chlorhexidine group vs 53.8% in the alcoholic povidone-iodine group).

Overall, 261 patients (11.5%) experienced hematoma with no difference between both groups. Noninfectious side effects attributable to local treatment: 18 patients (1.6%) in the alcoholic chlorhexidine group and 22 patients (1.9%) in the alcoholic povidone-iodine group (Table 2). Serious adverse events occurred in four patients (0.2%), 2 in each group. There were no chemical burns or surgical fires reported in either trial group.

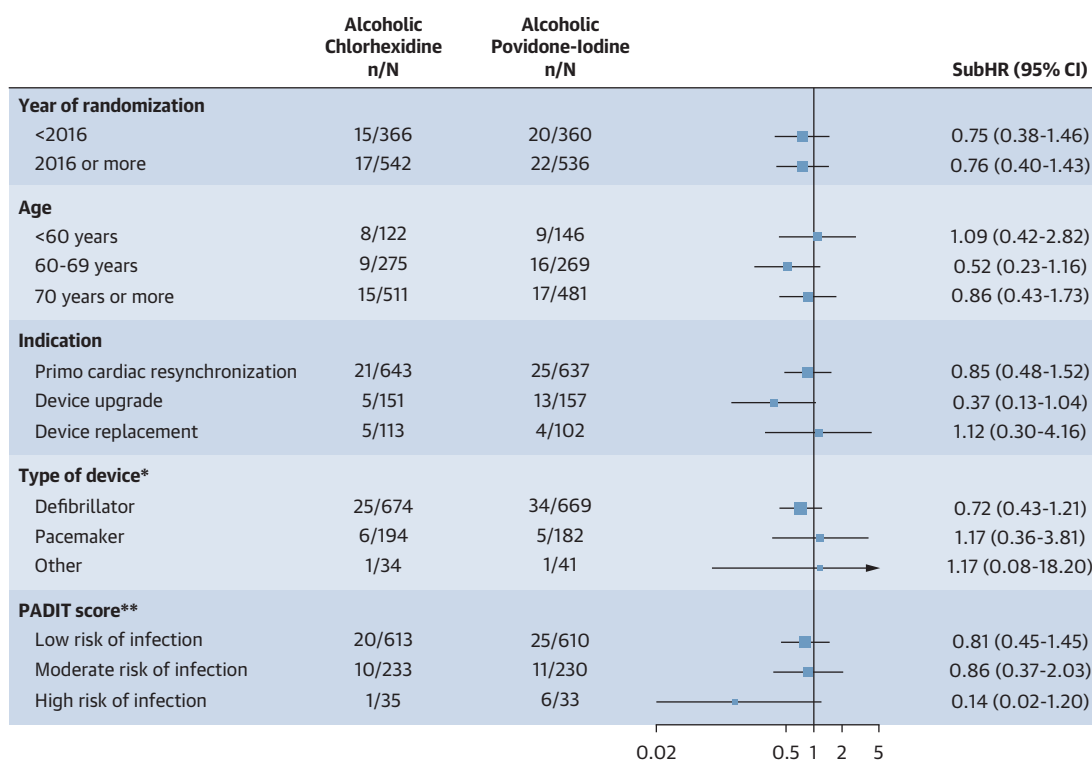
DISCUSSION

The CHLOVIS trial did not demonstrate a significant reduction in infection rates with 2% alcoholic chlorhexidine compared with alcoholic povidone-iodine, despite a 25% relative risk reduction observed with chlorhexidine. These results indicate that while a clinically meaningful benefit cannot be excluded, the CI also includes the possibility of no effect. Both antiseptic agents were associated with similar rates of infections and cardiovascular events. The results were consistent regardless of the type of device.

Safety outcomes were comparable between groups, with few major adverse events related to the antiseptic choice. This is the first large, randomized, multicenter trial, comparing 2 alcoholic antiseptic solutions—chlorhexidine vs povidone-iodine—for prevention of CIED infections. The study confirms that CRT device implantation carries a high infection risk (3.4% overall). Although chlorhexidine-alcohol showed a lower cumulative infection incidence (2.9%) than povidone-iodine-alcohol (3.9%), this difference was not statistically significant, reflecting only a trend towards risk reduction. These findings support the hypothesis that the alcohol component may play a more crucial role than the active antiseptic agent itself.

During the last decade, new advances have been created and statements have been published addressing CIED infections on what prevention to use (Central Illustration).^{1-5,19} Despite many improvements in CIEDs prevention and approaches, the infectious risk remains still or increases significantly.^{5-9,15,16,19,20} For example, the 2011-2018 national health care administrative databases in the United States reported a total of 1,604,173 admissions for CIED implantations and 71,007 admissions

FIGURE 3 Forest Plot of Subgroups: Primary Outcome (Local or Systemic Infection in Relation to the Implantable Device) (Intention-to-Treat Population)



*Data unavailable for 69 patients. **Data unavailable for 122 patients. The sum of the subgroup counts does not correspond to the total number of patients included due to patients lost to follow-up or deceased (competing risk).

(4.4%) for CIED infections.²⁰ Many other reports illustrated that the infectious risk varies due to many factors such as illustrated in the BLISTER score and PADIT (Prevention of Arrhythmia Device Infection Trial) study.⁶⁻⁸ Scoring each patient with PADIT score represents a pragmatic approach. Thus, a total score of 0-4, 5-6, and ≥ 7 confers a CIED infection risk of <1%, 1%-2%, and 2.9%, respectively.^{6,7} Accordingly, due to the patients' status comorbidities, the number of leads use, the procedure duration, the device volume and a higher reintervention risk, CRT devices population represents the population at higher infectious risk.^{5,19-21}

Accordingly, considering that the per-procedure contamination represents CIED infections main mechanism, all our efforts should be concentrated on prevention particularly in the population exposed to the higher infectious risk such as patients needing a CRT device, which has been identified as carrying a higher risk of infection compared with other CIED types.^{5,9,10,13,21} No recent information was provided regarding preprocedural intravenous antibiotic therapy, which remains based on international

consensus.^{4,11,17} Prophylactic systemic antibiotics that cover at least the *Staphylococcus aureus* species are required, consisting mainly of cefazolin (1-2 g) within 1 hour of incision.^{11,17} The routine use of postoperative antibiotics was not supported by recently published data.⁶ Concerning local antiseptic administration for preventing skin infection upon device implantation, randomized studies are still lacking, and the results are extrapolated from those reported with other surgical approaches.¹² Although little data from retrospective nonrandomized trials or registries specifically related to CIED implantation procedures exist, the antiseptic agent choice for skin preparation remains of debate.^{7,14,22-27} The lack of directly applicable research and the conflicting results in other surgical populations have contributed to ongoing uncertainty among surgeons with respect to the most appropriate type of antiseptic in such cases.^{12,13} Accordingly, 57.8% of centers reported using a povidone-iodine solution as skin antiseptic, while the use of chlorhexidine was reported in 42.2% of centers.²⁶ However, over the time, the practices moved towards the use of the chlorhexidine-alcohol

CENTRAL ILLUSTRATION Chlorhexidine vs Povidone-Iodine Alcohol Solutions for Cardiac Implantable Electronic Devices

Both chlorhexidine and povidone-iodine antiseptic strategies are reasonable when implanting a CIED. **The alcohol is likely the key factor in infection prevention.**

The largest randomized study to date comparing local antiseptic strategies for cardiac device implantation—a clinically meaningful and unresolved question that matters greatly to patients, electrophysiologists, heart failure specialists, and procedural cardiologists.

TRIAL DESIGN AND POPULATION

21 centers
across France

2,272 patients
undergoing CRT

Randomized 1:1
to chlorhexidine (n = 1,143) or
povidone-iodine (n = 1,129)

**Randomization April 2013
to December 2018**

RESULTS**PRIMARY OUTCOME**

**Device-related
infections at
24 months**

CHX 2.9%
POV-I 3.9%
(–25%, P = NS)

SECONDARY OUTCOMES

Major CV events 31.5% 31.3%

Non-infectious side effects 1.6% 1.9%

CRT implantation carries a high infection risk (3.4% overall).

The CHLOVIS trial did not demonstrate a significant reduction in infection rates with 2% alcoholic chlorhexidine compared to alcoholic povidone-iodine.

Da Costa A, et al. JACC. 2026;87(2):200–211.

solution and, for example, in the PADIT trial, chlorhexidine-alcohol solution was used for antiseptic skin preparation in 92.9% of cases (n = 19,603) essentially in North America.⁷ To our knowledge, no randomized study has yet been conducted comparing a chlorhexidine-alcohol solution vs a povidone-iodine-alcohol solution in the setting of CIED infection prevention. Darouiche et al¹² published a landmark randomized study comparing chlorhexidine-alcohol vs aqueous povidone-iodine in general, abdominal, and nonabdominal (such as thoracic, gynecologic, and urologic) surgery. Chlorhexidine-alcohol was significantly more protective than aqueous povidone-iodine against both superficial incisional infections (4.2% vs 8.6%; $P = 0.008$) and deep incisional infections (1% vs 3%; $P = 0.05$), but not against organ-space infections

(4.4% vs 4.5%).¹² Aqueous povidone-iodine cannot be equated with povidone-iodine-alcohol solution. Thus, general recommendations in the literature suggest that the “active ingredient” of skin antiseptics could potentially be the alcohol solution due to its antimicrobial properties.²⁴ More recently, the debate was revived by the PREP-IT (Program of Randomized Trials to Evaluate Preoperative Antiseptic Skin Solutions in Orthopedic Trauma) study demonstrating that patients with closed extremity fractures, skin antisepsis with iodine povacrylex alcohol resulted in fewer surgical-site infections than antisepsis with chlorhexidine gluconate in alcohol (2.4% vs 3.3%; OR: 0.74; $P = 0.049$).²⁸ By contrast, the risk of surgical-site infection did not differ significantly between the 2 trial groups in the open-fracture population.²⁸ The iodophor that was used

in PREP-IT trial differs from povidone iodine.²⁸ Iodine povacrylex is a novel iodophor that is available in alcohol and distinguished by its copolymer, which allows povacrylex to prevent iodine inactivation by organic matter.²⁸

Our trial has several strengths. It represents the largest prospective recruitment in the CRT population to date. Infection rates remain a current concern in 2025, as we observed 3.9% infections in the alcoholic povidone-iodine group, close to the 4% rate anticipated in the protocol. The enrollment of patients from 21 geographically diverse hospitals across France enhances the generalizability of the results. This pragmatic trial compared the 2 most commonly used alcohol-based antiseptic solutions worldwide for CIEDs. The study was conducted in high-volume centers with experienced operators, including 13 academic medical centers (62%) and 8 general hospitals (38%), which further supports the applicability of the findings to routine clinical practice. Last, the results remain highly relevant to current clinical settings. Thus, the use of the 2% alcoholic chlorhexidine is much simpler to apply with a total application time of 2 minutes without prior preparation by the nurse compared with around 10 minutes or more for the alcoholic povidone-iodine solution.

STUDY LIMITATIONS. First, the study was open-label, but blinded adjudication of outcomes was performed to minimize assessment bias. Second, the COVID-19 pandemic complicated the completion of follow-up for some of the last enrolled patients; however, the rate of patients lost to follow-up remained low (2.9%). Third, although follow-up was limited to 2 years and infection risk may persist beyond this period—raising concerns about reduced statistical power—most events in the CHLOVIS trial occurred within the first 12 months, suggesting this limitation likely had minimal impact. Fourth, the long recruitment period involved patients managed under evolving clinical practices. Nonetheless, changes in practice before and after 2016 (PARADIGM-HF [Prospective Comparison of ARNI with ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure] study) did not affect the treatment effect (Figure 3). Fifth, since the design study was elaborated, new preventive actions have been developed such as antibiotic-eluting envelope use. It is interesting to note that the treatment effect is comparable to that seen in PADIT, but one-half of that seen in WRAP IT (World-wide Randomized

Antibiotic Envelope Infection Prevention Trial), likely because of lack of potent treatment within the pocket. Thus, the WRAP-IT investigators reported a 40% reduction in major infections at 12 months, and the infectious risk reduction in this population makes it even more difficult to construct a new powered study in the future.²⁹ In our study, no patient received an antibiotic envelope because WRAP-IT was released on March 17, 2019, whereas our last inclusion occurred on December 18, 2018. Moreover, the envelope was only officially added to the list of reimbursable products and services in France on April 16, 2021, well after the end of inclusions. Sixth, the PADIT risk score was not available for all patients, as the study began before its development, and not all the necessary variables were collected. These deficiencies in available data regarding the PADIT risk score due to timing of the study and the "need" to include antimicrobial envelopes in the high-risk patients would represent ethical concerns today about in subsequent prevention studies. "Seventh, no specific assessment of adherence to the antiseptic protocol was conducted, although it is reasonable to assume that compliance was comparable across groups. Moreover, the methodology used is consistently validated in protocols established by hygienists in French hospitals." Finally, although the 25% reduction in infections with chlorhexidine may have been underestimated by the study, the infection rate observed in the povidone-iodine-alcohol group (3.9% after 2 years) matched to the rate anticipated in the initial sample size calculation (4%). Using these observed rates as assumptions for a future trial, and assuming 80% statistical power with a 2-sided type I error of 5%, approximately 5,440 patients per group would be required.

CONCLUSIONS

The study did not demonstrate a significant reduction in infection rates with alcoholic chlorhexidine compared with alcoholic povidone-iodine in patients undergoing cardiac resynchronization device implantation. Both antiseptics showed similar rates of major infectious and noninfectious events. Further research is needed to optimize infection prevention strategies in this population.

DATA AVAILABILITY The data underlying this article will be made available by the corresponding author upon reasonable request. Deidentified individual participant data and relevant study documents

(including the protocol and statistical analysis plan) may be shared with qualified investigators for legitimate research purposes, pending approval and in accordance with institutional and regulatory requirements.

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REFERENCES

- Goette A, Sommer P. Infections of cardiac implantable electronic devices: still a cause of high mortality. *Europace*. 2021;23:iv1–iv2.
- Han HC, Hawkins NM, Pearman CM, Birnie DH, Krahn AD. Epidemiology of cardiac implantable electronic device infections: incidence and risk factors. *Europace*. 2021;23:iv3–iv10.
- Baddour LM, Esquer Garrigos Z, Rizwan Sohail M, et al. Update on cardiovascular implantable electronic device infections and their prevention, diagnosis, and management: a scientific statement from the American Heart Association: endorsed by the International Society for Cardiovascular Infectious Diseases. *Circulation*. 2024;149(2):e201–e216.
- Blomstrom-Lundqvist C, Traykov V, Erba PA, et al, ESC Scientific Document Group. European Heart Rhythm Association (EHRA) international consensus document on how to prevent, diagnose, and treat cardiac implantable electronic device infections—endorsed by the Heart Rhythm Society (HRS), the Asia Pacific Heart Rhythm Society (APHRS), the Latin American Heart Rhythm Society (LAHRS), International Society for Cardiovascular Infectious Diseases (ISCVID) and the European Society of Clinical Microbiology and Infectious Diseases (ESCMID) in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). *Europace*. 2020;22:515–549.
- Køber L, Thune JJ, Nielsen JC, et al, for the DANISH Investigators. Defibrillator implantation in patients with nonischemic systolic heart failure. *N Engl J Med*. 2016;375:1221–1230. <https://doi.org/10.1056/NEJMoa1608029>
- Birnie DH, Wang J, Alings M, et al. Risk factors for infections involving cardiac implanted electronic devices. *J Am Coll Cardiol*. 2019;74:2845–2854.
- Krahn AD, Longtin Y, Philippon F, et al. Prevention of arrhythmia device infection trial. *J Am Coll Cardiol*. 2018;72:3098–3109.
- Maclean E, Mahatani K, Honarbakhsh S, et al. The BLISTER score: a novel externally validated tool for predicting CIEDI and its cost-utility implications for antimicrobial envelope use. *Circ Arrhythm Electrophysiol*. 2024;17(3):e012446. <https://doi.org/10.1161/CIRCEP.123.012446>
- Romeyer-Bouchard C, Da Costa A, Dauphinot V, et al. Prevalence and risk factors related to infections of cardiac resynchronization therapy devices. *Eur Heart J*. 2010;31:203–210.
- Da Costa A, Lelievre H, Kirkorian G, et al. Role of the preaxillary flora in pacemaker infections: a prospective study. *Circulation*. 1998;97(18):1791–1795. <https://doi.org/10.1161/01.cir.97.18.1791>
- Da Costa A, Kirkorian G, Cucherat M, et al. Antibiotic prophylaxis for permanent pacemaker implantation: a meta-analysis. *Circulation*. 1998;97(18):1796–1801.
- Darouiche RO, Wall MJ, Itani KMF, et al. Chlorhexidine-alcohol versus povidone iodine for surgical site antisepsis. *New Engl J Med*. 2010;362:18–26.
- Da Costa A, Benali K. Cardiac implantable electronic device infection: prevention is better than cure. *Trends Cardiovasc Med*. 2025;35(5):317–319.
- Da Costa A, Tulane C, Dauphinot V, et al. Preoperative skin antiseptics for prevention of cardiac implantable electronic device infections: a historical-controlled interventional trial comparing aqueous against alcoholic povidone-iodine solutions. *Europace J*. 2015;17(7):1092–1098.
- Tang AS, Wells GA, Talajic M, et al, Resynchronization-Defibrillation for Ambulatory Heart Failure Trial Investigators. Cardiac-resynchronization therapy for mild-to-moderate heart failure. *N Engl J Med*. 2010;363(25):2385–2395.
- Leclercq C, Cazeau S, Lellouche D, et al. Upgrading from single chamber right ventricular to biventricular pacing in permanently paced patients with worsening heart failure: the RD-CHF Study. *Pacing Clin Electrophysiol*. 2007;30(Suppl 1):S23–S30.
- de Oliveira JC, Martinelli M, Nishioka SA, et al. Efficacy of antibiotic prophylaxis before the implantation of pacemakers and cardioverter-defibrillators: results of a large, prospective, randomized, double-blinded, placebo-controlled trial. *Circ Arrhythm Electrophysiol*. 2009;2(1):29–34.
- Delgado V, Ajmone Marsan N, de Waha S, et al, ESC Scientific Document Group. 2023 ESC Guidelines for the management of endocarditis. *Eur Heart J*. 2023;44(39):3948–4042. <https://doi.org/10.1093/eurheartj/ehad193>
- Fine JP, Gray RJ. A proportional hazards model for the subdistribution of a competing risk. *J Am Stat Assoc*. 1999;94:496.
- Modi V, Shah K, Ferraro B, et al. Cardiac implantable electronic device implantation and device-related infection. *Europace*. 2023;25:1–10.
- Olsen T, Jørgensen OD, Cosedis Nielsen J, et al. Risk factors for cardiac implantable electronic device infections: a nationwide Danish study. *Eur Heart J*. 2022;43(47):4946–4956.
- Quintar M, Zardkoobi O, Hammadah M, et al. The impact of changing antiseptic skin preparation agent used for cardiac implantable electronic device (CIED) procedures on the risk of infection. *Pacing Clin Electrophysiol*. 2015;38:240–246.
- Da Costa A, Ovache L, Chellali S, et al. Preoperative skin antiseptics for the prevention of cardiac implantable electronic device infections: chlorhexidine-alcohol versus povidone-iodine-alcohol. *J Interv Card Electrophysiol*. 2025;68(8):1563–1572. <https://doi.org/10.1007/s10840-025-02049-0>
- Dumville JC, McFarlane E, Edwards P, Lipp A, Holmes A. Preoperative skin antiseptics for preventing surgical wound infections after clean surgery. *Cochrane Database Syst Rev*. 2013;3:1–64.
- Chen HC, Chen MC, Chen YL, Tsai TH, Pan KL, Lin YS. Bundled preparation of skin antiseptics decreases the risk of cardiac implantable electronic device-related infection. *Europace*. 2016;18(6):858–867.

26. Bongiorni G, Marinskis G, Lip G, Svendsen JH, Dobreanu D, Blomström-Lundqvist C, Scientific Initiative Committee, European Heart Rhythm Association. How European centres diagnose, treat and prevent CIED infections: results of an European Heart Rhythm Association survey. *Europace J*. 2012;14:1666-1669. <https://doi.org/10.1093/europace/eus350>

27. Uslan DZ, Gleva MJ, Warren DK, et al. Cardiovascular implantable electronic device replacement infections and prevention: results

from the REPLACE Registry. *Pacing Clin Electrophysiol*. 2012;35:81-87.

28. PREP-IT Investigators, Sprague S, Slobogean G, Wells JL, et al. Skin antisepsis before surgical fixation of extremity fractures. *N Engl J Med*. 2024;390:409-420.

29. Tarakji KG, Mittal S, Kennergren C, et al. Antibacterial envelope to prevent cardiac implantable device infection. *N Engl J Med*. 2019;380:1895-1905.

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APPENDIX For a complete list of the CHLOVIS Study Investigators and an expanded Methods section, please see the online version of this paper.